

Geographic Variation in Operative Management of Adhesive Small Bowel Obstruction.

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Author Contributions:

S.P.C. contributed to the conception and design of the study, analysis and interpretation of data, drafting and revision of the article, and finalized the version being submitted. D.M.K. contributed to the conception and design of the study, analysis and interpretation of data, and drafting and revision of the article. N.T.M., P.R.M., and J.W.M. contributed to the conception and design of the study, analysis and interpretation of data and revision of the article for important intellectual content. A.D.H. contributed to the conception and design of the study, acquisition of data, analysis and interpretation of data and revision of the article for important intellectual content.

Background:

Variation in surgical management exists nationally. We hypothesize that geographic variation exists in adhesive small bowel obstruction (aSBO) management.

Materials and Methods:

A retrospective analysis of a national commercial insurance claims database (MarketScan) sample (2017–2019) was performed in adults with hospital admission due to aSBO. Geographic variation in rates of surgical intervention for aSBO was evaluated by state and compared to a risk adjusted national baseline using a Bayesian spatial rates Poisson regression model. For individual-level analysis, patients were identified in 2018, with 365-day look-back and follow up periods. Logistic regression was performed for individual-level predictors of operative intervention for aSBO.

Results:

Two thousand one hundred forty-five patients were included. State level analysis revealed rates of operative intervention for aSBO were significantly higher in Missouri and lower in Florida. On individual-level analysis, age ($p < 0.01$) and male sex ($p < 0.03$), but not comorbidity profile or prior aSBO, were negatively associated with undergoing operative management for aSBO. Patients presenting in 2018 with a history of admission for aSBO the year prior experienced a five-fold increase in odds of re-presentation (OR: 5.4, 95% CI: 3.1–9.6) in 2019. Patients who received an operation for aSBO in 2018 reduced the odds of readmission in the next year by 77% (OR: 0.23, 95% CI: 0.1–0.5). The volume of operations performed within a state did not influence readmission.

Conclusions:

Surgical management of aSBO varies across the continental USA. Operative intervention is associated with decreased rates of re-presentation in the following year. These data highlight a critical need for standardized guidelines for EGS patients.

Introduction

Small bowel obstruction due to adhesions (aSBO) is a common diagnosis within the Emergency General Surgery (EGS) patient population, resulting in over 250,000 admissions annually, and is a major driver of hospital costs and readmissions.^{1–3} The gold standard for treatment is surgical division of the intra-abdominal scar tissue (adhesiolysis).⁴ In patients presenting with inadequate blood flow to the bowel or bowel perforation due to aSBO, immediate surgical management is imperative. However, a non-operative approach with bowel rest and symptom management is frequently undertaken in the absence of these factors with successful resolution of aSBO in approximately 70% of cases.⁴ In this group

of patients, the timing and rationale to proceed with surgical care is variable among surgeons and hospitals, as operation is considered a failure of medical management.^{4,5}

Across surgical specialties, variation in performance of operative procedures is common by region of the country.^{6–8} Previous work has shown that geographic residence is often more predictive of undergoing an operation than clinical context.⁸ Other factors, such as patient demand, physician beliefs, and incorporation of patient preferences, beyond differences in illness burden and practice guidelines, are potential drivers for performance of an operation.^{8,9} Recognizing the disease heterogeneity of aSBO, we hypothesize that geographic variation exists across the continental United States in the performance of operative management. Furthermore, these differences in practice may affect readmission. To analyze this, we retrospectively compared rates of surgical intervention for aSBO by state using a national commercial insurance claims database covering >25 million individuals aged <65 years.

Materials and methods

Setting and Design

We performed a population-based retrospective cohort study to evaluate the state-level variation in management and outcomes of patients with aSBO in the continental United States of America (USA).

Data Source

Data were collected through the IBM MarketScan Research Databases, a national commercial insurance claims database covering >25 million individuals aged <65 years, employed by >300 self-insured employers and representing >25 US health plans.¹⁰ Health information for employees (primary insured individuals) and family members, as well as dependents, are included in 4 databases: demographic records, inpatient medical records, outpatient medical records, and outpatient prescription records. Specific data points included are age, sex, index condition, procedures performed, comorbid conditions, prescription medications, discharge status, geography, and provider information (i.e., hospitals, physicians). Mortality is not an available data point in the database. The data have been previously collected and de-identified, thus are Health Insurance Portability and Accountability Act (HIPAA) compliant. Institutional Review Board approval was additionally obtained (IRB00075966).

Cohort

We identified a cohort of patients aged 18–64 years who were admitted to hospital for aSBO from 2017–2019 in the continental USA. Patients were identified within the sample by the following International Classification of Diseases 10th edition (ICD-10) diagnosis codes for intestinal adhesions with obstruction (K56.50), intestinal adhesions with partial obstruction (K56.51), and intestinal adhesions with complete obstruction (K56.52). Patients were excluded who experienced bowel obstruction due to other causes or were lost to follow up due to death or changes in insurance coverage. The following were excluded ICD-10 diagnosis codes: K56.2 (diagnosis of volvulus); K40.x-K46.x (diagnosis of hernia); K56.0, K56.7 (diagnosis of paralytic ileus or ileus, unspecified); K56.1 (diagnosis of intussusception); K56.3 (diagnosis of gallstone ileus); K56.4 (diagnosis of impaction of intestine); K50, K51 (diagnosis of inflammatory bowel disease); C16–18.x C19, C20, C21.x, C26.0, C54.x, C56.x, C61.x, C67.x, C76.2, C76.3, C76.7, C76.8 (diagnosis of abdominal malignancy); C77.2, C78.4, C78.5, C78.6, C78.8, C79.0, C79.1, C79.6, C79.7, C79.8, C80.x (carcinomatosis). This method for identification of patients with aSBO is similarly described in the contemporary literature.^{11,12} Patients were uniquely identified in 2018 with a 1 year (365 days) look-back (pre-index) period in 2017 for comorbidity characterization. Patients with multiple hospitalizations were included as separate data points. Patients undergoing operative intervention for aSBO were identified by the following Current Procedural Terminology (CPT) codes: 44180 (laparoscopy, surgical with enterolysis), 44005 (enterolysis, freeing of intestinal adhesion).

Outcomes

Our primary outcome measure is the rate of surgical intervention for aSBO by state in the continental USA. Secondary outcomes include rates of readmission in the following year after index presentation in 2018. We also explored the relationship between readmission in the following year in patients who underwent operation for aSBO and volume of aSBO operations performed in a state.

Covariates

We identified individual and state-level covariates that were potential confounders to include in our Poisson regression model. Individual covariates were pre-index patient characteristics observed the year (365 days) prior to admission in 2018 for aSBO such as age, sex, prior admission for aSBO and patient comorbidities from the weighted Elixhauser comorbidity categories.^{13,14} The weighted Elixhauser score is a validated modification of the 30 binomial variable Elixhauser comorbidity score into a single numeric score for the purposes of comparing administrative datasets. State-level covariates included state population, population density, number of surgeons per state, number of surgeons per 100,000 population, number of hospitals per state, geography of a hospital (urban versus rural), and number of Accreditation Council for Graduate Medical Education (ACGME) affiliated institutions per state to represent the number of surgical training programs per state.

Analytic Approach

Geographic variation in rates of surgical intervention for aSBO was evaluated by state and compared to a risk adjusted national baseline using a Bayesian spatial rates Poisson regression model.^{15,16} A probability of operation was estimated for each patient based on their pre-index characteristics and those probabilities were summed within a state to give the expected number. We computed posterior probabilities of the probability that a state operates more than expected. To both account for spatial dependence and help to stabilize estimates from states with small counts, we included a spatial random effect using a conditional autoregressive model.¹⁷ Similarly, Alaska and Hawaii were excluded due to small counts and the absence of a nearest neighbor for spatial correlation. Risk-adjusted outcomes were expressed as an observed-to-expected (O/E) operative rate ratio by state, as an application of previously described methodologies by the senior author.^{18,19} We plotted each state's standardized rate ratio and 95% credible interval, or certainty, after controlling for comorbidities to evaluate for significant differences. Logistic regression was performed for individual-level predictors of operative intervention for aSBO. We subsequently evaluated each state for rates of readmission the following year based upon operative volumes for aSBO.

We performed descriptive statistics for assessment of baseline characteristics, including means with standard deviations (SD) for continuous variables with categorical variables expressed as counts with percentages. Analyses were performed using R statistical software using the NIMBLE package and SAS Studio (SAS Institute, Cary, NC).²⁰ Statistical significance was set a priori at $p < 0.05$ (two-sided).

Results

Over the study period, a total of 2,145 patients, identified in 2018, were included in our sample. Overall, 23% ($n=500$) of patients were managed operatively. Demographics of operative versus non-operative patients are presented in Table 1. Patients managed with operation were younger ($p < 0.01$) and more likely to be female ($p < 0.03$). Comorbidity profile, evidence of prior aSBO and state-level covariates did not demonstrate a significant association with odds of operative intervention in our sample. A total 4% ($n=90$) of patients admitted with aSBO in 2018 were re-admitted the following year with this diagnosis. For each patient identified in 2018, a history of admission for aSBO in 2017 (within 365 days prior to 2018 admission), independent of management strategy at that time, experienced a five-fold increase in odds of readmission (OR: 5.4, 95% CI: 3.1–9.6) in 2019. However, patients who received an operation for aSBO in 2018 reduced the odds of readmission for aSBO in the next year by 77% (OR: 0.23, 95% CI: 0.1–0.5). The volume of operations performed within a state for aSBO did not influence readmission for patients who underwent surgical management of aSBO.

Rates of operative intervention for aSBO were significantly higher in Missouri and lower in Florida compared to a national baseline (Figure 1). Adjustment for comorbidity profile only demonstrated higher operative intervention in Missouri (Figure 2). At the state level, comorbidity profile was not influential over re-admission for aSBO (Figure 3). There were no significant differences evident in rates of re-presentation for states with more versus less aggressive strategies of operative management for aSBO (Figure 4).

Discussion

Regional variations in the care of surgical conditions is evident across both time and surgical subspecialties.^{6,8} Moreover, the odds of a patient undergoing an operation for a given indication may depend more on where they reside than on the clinical circumstance.⁸ Though this finding is well-established among elective surgical practices, regional variation remains to be defined in Emergency General Surgery patients, who represent a socio-economically diverse and clinically heterogeneous population. Using a US national administrative claims database (IBM MarketScan), we present data demonstrating for the first time differences in management of aSBO occurring at the state level that are incompletely explained by individual and state-level covariates. State-level analysis revealed rates of operative intervention for aSBO were significantly higher in Missouri and lower in Florida. On an individual level, patients who received an operation for aSBO in 2018 reduced the odds of readmission in the next year by 77% (OR: 0.23, 95% CI: 0.1–0.5).

Emergency General Surgery (EGS), as defined by the American Association for the Surgery of Trauma (AAST), is “any patient (inpatient or emergency department) requiring an emergency surgical evaluation (operative or non-operative) for diseases within the realm of general surgery as defined by the American Board of Surgery”.²¹ EGS accounts for approximately 3 million annual admissions at over 4700 hospitals in the USA, exceeding the incidence of diabetes, cancer and heart disease, with a cost of over \$28 billion for annual inpatient care.²¹ Small bowel obstruction due to adhesions (aSBO) is the 3rd most commonly evaluated EGS diagnosis and constitutes 1% of all hospitalizations in the USA.^{5,22} Regardless of this high incidence, there remains large variation in the implementation of operative management and the timing of operation during a given hospitalization.⁵

In general terms, regional variations in surgical practice may be broken down into 3 categories: patient demand, physician beliefs, and incorporation of patient preferences.⁸ Patient demand is further sub-categorized by the true incidence of surgically-treatable disease, patient willingness to undergo operation and subclinical disease detection, which is not possible in aSBO.⁸ Previous studies have shown a linear correlation between the incidence of certain disease entities (i.e. hip fracture) and operation.²³ We explored this hypothesis by evaluating the incidence of aSBO by state (Supplemental Figure 5). By contrast, Florida, a state with a statistically lower observed-to-expected ratio of operations for aSBO demonstrated among the highest rates of aSBO as an admitting diagnosis. Interestingly, this observation is not explained by individual comorbidity profile (Figure 2), though increasing population age may play a significant role.

Patient willingness to undergo an operation is multifactorial and significant gaps in understanding exist regarding patient preferences in the emergent surgical setting.²⁴ Previous observations from the orthopedic literature suggest that patients residing in high-rate areas for a given procedure are more willing to undergo operation.²⁵ However, in the present study, states with higher rates of presentation for aSBO do not necessarily have higher operative rates (Supplemental Figure 6). While this finding does not isolate patient willingness as a single predictor, it suggests that this is not a major driver in state-level variation. In further support, multiple prior studies have demonstrated that patient preferences for care does not vary greatly across regions and patient demand remains relatively unimportant to regional variation in health care expenditure.^{26,27}

A recent nationwide survey study by Cutler et al assessed the impact of physician (supply-side) beliefs and patient (demand-side) preferences in Medicare expenditures.⁹ While patient preferences played a modest predictive role in spending patterns, local physician beliefs had a much stronger role in explaining expenditures. Though organizational factors, such as colleague accommodation, financial considerations

and malpractice concerns were reported as influencers in physician decision-making, the largest degree in variation was explained by physician beliefs regarding therapeutic efficacy of a particular treatment. Furthermore, surveyed physicians had strongly differing opinions about how to treat the same patient that were often inconsistent with evidence-based society guidelines for optimal care.⁹ Insight into the contribution of physician beliefs, while not available in the present dataset, will likely explain additional regional operative variation in future work.

Recent analyses have shown that, independent of hospital type and volume of aSBO admissions, patients operated earlier in their admission (i.e., day 0–1) had lower likelihoods of serious complications, bowel resections and death.⁵ Furthermore, operative management at the time of first aSBO admission reduces the risk of recurrence and increases long-term survival, though the disease trajectory may be altered by operation at any readmission for aSBO.^{11,12,28} In the present study, 23% of patients underwent operation for aSBO. Patients with a history of admission for aSBO, independent of management strategy, experienced a five-fold increase in odds of readmission (OR: 5.4, 95% CI: 3.1–9.6), while operative intervention decreased the likelihood of readmission for aSBO by 77% (OR: 0.23, 95% CI: 0.1–0.5). Our findings are consistent with contemporary population-based analyses, contributing to the validity and generalizability of these data in our national sample.

Interestingly, we found that states performing a higher volume of operations for aSBO do not necessarily have lower rates of readmission (Figure 4). This corresponds with the observation that regions delivering a higher volume of care may not have better outcomes for patients.²⁹ Moreover, physicians in higher spending settings are more likely to perform discretionary interventions where judgement is involved.³⁰ Moving forward, improvements in standards of care will join the definitions of aSBO severity with outcomes in a robust way for generation of optimal guidelines.

The major strengths of this study are the population-based observations drawn from a national commercial insurance claims database (IBM MarketScan) covering >25 million individuals aged <65 years with longitudinal follow-up. There are limitations to this study that are inherent to this and other databases in the USA versus other single-payer systems. Elderly patients ≥ 65 years of age and those insured by Medicaid are not included, creating a selection bias toward the younger and employed population. We are unable to fully characterize the number of bowel obstructions per patient, given the limited look-back period. The follow up period is also limited, restricting the scope of long-term benefits for operative intervention in aSBO. However, prior studies suggest the greatest risk of obstruction exists within the first 2 years after operation.³¹ Given the administrative nature of the data, some granularity including physiologic data, race and ethnicity and mortality are not included.

Conclusions

The present study highlights state-level variation in the performance of surgical management for aSBO apart from traditionally known individual and state-level covariates. Furthermore, this practice variation emphasizes the critical need for standardization in the care of EGS patients.³² Guidelines developed by surgical societies reflect the evolution of contemporary understanding. The goal of this and other similar investigations is to reduce the “grey area of clinical discretion” through population-based analyses and robust definitions.⁸ The patient outcomes of the present study reflect prior population-based analyses and show for the first time that practice variation exists on the state level across the continental USA. Future directions may include a larger number of aSBO cases for comparing differences by region and estimation of readmission rate. Additionally, evaluation into less commonly evaluated predictors, such as incorporation of patient preferences, shared decision-making and goal-concordant care may provide further insight into observed variation.^{33,34}

Supplementary Material

Acknowledgements

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Declarations of interest: None

References

Geographic variation heat map showing surgical intervention for aSBO evaluated by state compared to a risk-adjusted national baseline. Significant differences from baseline are shown in Missouri and Florida (Posterior Mean Log Standardized Rate Ratio: 2018).

Forrest plot showing rates of surgical intervention for aSBO by state in 2018 compared to a comorbidity risk-adjusted national baseline (Standardized Rate Ratio estimates with 95% credible intervals).

Forrest plot showing readmission for aSBO by state in 2019 compared to a comorbidity risk-adjusted national baseline (Standardized Rate Ratio estimates with 95% credible intervals).

Geographic variation heat map showing readmission for aSBO evaluated by state compared to a risk-adjusted national baseline. (Posterior Mean Log Standardized Rate Ratio: 2018).

Demographics of patients who underwent operative versus non-operative intervention.